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Many-body perturbation theory vs. density functional theory: a systematic benchmark for band gaps of solids

Max Großmann^{1,2}✉, Marc Thieme^{1,2}, Malte Grunert¹ & Erich Runge¹

We benchmark many-body perturbation theory against density functional theory (DFT) for the band gaps of solids. We systematically compare four GW variants— G_0W_0 using the Godby-Needs plasmon-pole approximation (G_0W_0 -PPA), full-frequency quasiparticle G_0W_0 (QPG $_0W_0$), full-frequency quasiparticle self-consistent GW (QSGW), and QSGW augmented with vertex corrections in W (QSG $\hat{W}$)—against the currently best-performing and popular density functionals mBJ and HSE06. Our results show that G_0W_0 -PPA calculations offer only a marginal accuracy gain over the best DFT methods, however, at a higher cost. Replacing the PPA with a full-frequency integration of the dielectric screening improves the predictions dramatically, almost matching the accuracy of the QSG $\hat{W}$. The QSGW removes starting-point bias, but systematically overestimates experimental gaps by about 15%. Adding vertex corrections to the screened Coulomb interaction, i.e., performing a QSG $\hat{W}$ calculation, eliminates the overestimation, producing band gaps that are so accurate that they even reliably flag questionable experimental measurements.

Ever since the dawn of condensed matter physics, the band gap of semiconductors and insulators has been one of the most important material properties for optical and optoelectronic applications¹. Despite its relative ease of measurement, accurately predicting the band gap from first principles remains a challenging task, especially because interpreting the Kohn-Sham (KS) gap from density functional theory (DFT), the workhorse of theoretical materials science, as the fundamental band gap leads to a systematic underestimation of the band gaps of solids².

A benchmark by Borlido et al.^{3,4} consisting of 472 non-magnetic materials, evaluating the performance of 21 DFT functionals, showed that functionals from the upper rungs of Perdew's "Jacob's ladder"⁵, such as meta-generalized gradient approximation (meta-GGA) functionals and hybrid functionals, can significantly reduce the systematic underestimation of the band gap. However, these improvements are often due to (semi-) empirical adjustments and not the result of a solid theoretical basis⁴.

In contrast to DFT, many-body perturbation theory (MBPT) offers a fundamentally different way to obtain accurate electronic structures of solids. Based on a rigorous diagrammatic expansion of the electron correlation, MBPT provides, in principle, a systematic way to improve accuracy by incorporating higher-order corrections^{6,7}. Due to methodological advances over the last decades, in particular different "flavors" and implementations of the so-called GW approximation to the electronic self-energy^{8–17}, and in part due to ever-increasing computational power, MBPT

has evolved from a niche method to an attractive practical tool for accurate calculations of band gaps of solids. However, we must note that MBPT, as the name implies, is a perturbative method typically starting from a DFT-derived electronic structure, and is therefore generally more expensive to perform than a DFT calculation.

Despite its strong theoretical foundation and the common belief that MBPT has superior accuracy compared to state-of-the-art DFT functionals, a systematic large-scale benchmark directly comparing MBPT with state-of-the-art DFT functionals for the band gap of solids remains to be performed.

The demand for a systematic benchmark is particularly pressing in light of recent trends in materials science: the rise of machine learning (ML) and the need for accurate datasets¹⁸. ML models, which have rapidly become indispensable for accelerating the discovery of new materials and the fast prediction of key properties ranging from the transition temperature of conventional superconductors^{19–21} and diffusion coefficients^{22,23} to optical spectra^{24–26}, depend on training data, typically generated by DFT. Given the limitations of these DFT datasets, there is a steadily growing concern about the reliability of ML predictions, especially as models become more sophisticated and accurate. A promising solution to these concerns is transfer learning^{27,28}, which requires a (small) dataset of high-fidelity data for model retraining. Therefore, a benchmark of MBPT against DFT is also invaluable for the ML part of the materials science community, as one needs to decide on

¹Institute of Physics and Institute of Micro- and Nanotechnologies, Technische Universität Ilmenau, Ilmenau, Germany. ²These authors contributed equally: Max Großmann, Marc Thieme. ✉e-mail: max.grossmann@tu-ilmenau.de

the method of choice when computing expensive datasets for transfer learning.

Against this backdrop, our work provides a systematic benchmark of MBPT against the best meta-GGA and hybrid DFT functionals for calculating the band gap of solids, and answers the central question: “How well do MBPT calculations perform relative to the best available DFT functionals when predicting the band gaps of solids?”

The basis for our MBPT benchmark is the earlier DFT benchmark of Borlido et al.^{3,4}. We have adopted their extensive dataset of experimental band gaps for 472 non-magnetic semiconductors and insulators, using experimental crystal structures and geometries from the Inorganic Crystal Structure Database (ICSD)^{29–31} to facilitate direct a comparison. For details on the curation of the experimental data, as well as a detailed analysis of the dataset in terms of contained elements and band gap distribution, we refer to the original publications^{3,4}. The dataset of Borlido et al.^{3,4} includes not only experimental band gaps, but also band gaps calculated using a wide range of 21 DFT functionals, from which we have selected the best performing meta-GGA functional (mBJ)³², and the best performing hybrid functional (HSE06)^{33,34} for comparison. The selected functionals not only present the upper rungs of Perdew’s “Jacob’s ladder”, ensuring a broad and representative coverage, but also include one of the most widely used functionals in condensed matter physics, HSE06.

As the MBPT method of choice, we used the GW approximation, the arguably most successful and widely used MBPT method³⁵. Because the GW approximation exists in numerous variants, implementations, and workflows, it is impractical to cover every published variant exhaustively. Consequently, we have limited ourselves to a strategically chosen subset of four methods: (i) One-shot GW, so-called G_0W_0 calculations, utilizing the Godby-Needs plasmon-pole approximation (PPA)³⁶. (ii) Full-frequency quasiparticle G_0W_0 (QPG $_0W_0$) calculations. (iii) Full-frequency quasiparticle self-consistent GW (QSGW) calculations. (iv) QSGW calculations with vertex corrections in the screened Coulomb interaction W (QSG $\hat{W}$). Method (ii), i.e., a QPG $_0W_0$ calculation, is equivalent to stopping after a single iteration of method (iii), i.e., a QSGW calculation.

All calculations were automated using custom, in-house workflows to ensure reproducible results, see Methods. The workflow codes and generated data can be found in the Code and data availability statement. The starting point for all calculations was an LDA³⁷ DFT calculation. For method (i), we also repeated all G_0W_0 calculations starting from a PBE³⁷ DFT calculation to investigate the influence of the starting point. The DFT calculations in method (i) are based on norm-conserving pseudopotentials using a plane-wave basis, while those in methods (ii–iv) are all-electron calculations using a linear muffin-tin orbital (LMTO) basis. Calculations for (i) were performed using Quantum ESPRESSO^{38,39} and Yambo^{40,41}. Those for (ii–iv) were performed using the Questaal code^{10,17,42}. Details regarding the workflows, convergence parameters, and other computational choices are reported in the Methods section.

The four GW schemes were selected to represent a hierarchy of computational cost, physical rigor, and methodological maturity in MBPT. Method (i) is a widely used and comparatively inexpensive variant of the GW approximation, implemented in most plane-wave pseudopotential MBPT codes^{40,41,43–45}. Here, the quasiparticle (QP) energies ϵ_i^{QP} are calculated from the KS energies ϵ_i^{KS} via a linearized QP equation:

$$\epsilon_i^{\text{QP}} = \epsilon_i^{\text{KS}} + Z_i \langle \phi_i^{\text{KS}} | (\Sigma(\epsilon_i^{\text{KS}}) - V_{\text{xc}}^{\text{KS}}) | \phi_i^{\text{KS}} \rangle \quad (1)$$

neglecting the off-diagonal elements of the self-energy Σ . Here, $Z_i^{-1} = 1 - \langle \phi_i^{\text{KS}} | d\Sigma/d\epsilon |_{\epsilon_i^{\text{KS}}} | \phi_i^{\text{KS}} \rangle$ is the renormalization factor, $V_{\text{xc}}^{\text{KS}}$ is the KS exchange-correlation potential, and $|\phi_i^{\text{KS}}\rangle$ are KS states⁷. Methods (ii–iv) “quasiparticleize” the energy-dependent Σ by constructing a static Hermitian potential from it⁴⁶

$$\Sigma_0 = \frac{1}{2} \sum_{ij} |\psi_i\rangle \left\{ \text{Re}[\Sigma(\epsilon_i)]_{ij} + \text{Re}[\Sigma(\epsilon_j)]_{ij} \right\} \langle \psi_j| \quad (2)$$

replacing $V_{\text{xc}}^{\text{KS}}$. QP energies and wavefunctions are then obtained by solving the resulting effective QP equations, in which $V_{\text{xc}}^{\text{KS}}$ is replaced by Σ_0 . In Eq. (2) $|\psi_i\rangle$ are single-particle states of the resulting effective QP equations. In QPG $_0W_0$ calculations, they coincide with KS states, whereas in QSGW calculations (beyond the first iteration), they are QP states. Through Eq. (2), the off-diagonal elements of Σ are included, which turn out to be necessary to obtain correct band topologies⁴⁶, since the LDA incorrectly orders bands for, e.g., simple narrow-gap semiconductors like InN¹⁷ and PbTe⁴². The GW variant (ii) uses an all-electron starting point and replaces the PPA of the frequency dependence of the dielectric screening with a numerical integration that captures its exact frequency dependence. This is referred to as a full-frequency GW calculation and is also used in methods (iii) and (iv). The self-consistent QPGW scheme (iii), the QSGW, further eliminates the dependence on the initial DFT reference by iterating the quasiparticle Eq. (2) to convergence, i.e., until $\Sigma_0^{\text{in}} = \Sigma_0^{\text{out}42}$. In detail, one performs the following loop¹²:

$$\dots \rightarrow \Sigma_0 \rightarrow G_0 \rightarrow \Sigma = iG_0W[G_0] \rightarrow \Sigma_0 \rightarrow \dots \quad (3)$$

Here, Σ_0 is again obtained from Eq. (2) and G_0 is obtained from the solution of the effective QP equations, in which $V_{\text{xc}}^{\text{KS}}$ is replaced by Σ_0 . Finally, method (iv), the QSG $\hat{W}$ introduced in the seminal work by Cunningham et al.¹⁷, was chosen because it is, to our knowledge, the most advanced practical GW method. It has been found to produce extremely accurate band gaps for 43 more or less well-understood semiconductors, as well as for some strongly correlated antiferromagnetic oxides. The QSG $\hat{W}$ method accomplishes this by incorporating vertex corrections into the screened Coulomb interaction W . In particular, vertex corrections that augment dielectric screening are obtained by solving the Bethe-Salpeter equation (BSE) explicitly, thereby incorporating excitonic effects. In the employed Questaal implementation¹⁷, the BSE is solved for every momentum-transfer vector $\mathbf{q}$ in the $\mathbf{k}$ -point grid explicitly, using the usual static approximation of the BSE kernel in the Tamm-Dancoff approximation (TDA). Consequently, each QSG $\hat{W}$ iteration requires the computation of the full excitonic dispersion, resulting in a significant increase in computational cost compared to a QSGW calculation. At this point, we note that alternative implementations of vertex corrections in GW calculations already exist. For example, refs. 9,47,48 included ladder diagrams through an effective, nonlocal, static kernel constructed within time-dependent DFT to mimic the BSE. Furthermore, Tal et al.⁴⁹ proposed quasiparticle self-consistent schemes with vertex corrections included in W only, similar to Cunningham et al.¹⁷, as well as in both W and Σ . It is important to note that the notations used by Cunningham et al.¹⁷ and Tal et al.⁴⁹ to label their vertex-corrected QSGW methods are not consistent with each other and thus potentially confusing. Specifically, in the notation of Tal et al., QSG $\hat{W}$ denotes vertex corrections in both W and Σ , while QSG $\tilde{W}$ denotes vertex corrections only in W . In contrast, Cunningham et al.¹⁷ use QSG $\hat{W}$ for vertex corrections in W only, i.e., without vertex corrections in Σ . For clarity and consistency with the Questaal code employed in our work, we adopt the Cunningham convention and emphasize that a unified community standard would be highly desirable. In addition to these alternative methods, Kutepov^{15,50,51} proposed several schemes for self-consistently solving Hedin’s equations, including vertex corrections, which improve results for materials exhibiting complex electronic correlations beyond those obtained with the QSG $\hat{W}$ method. However, from a practical perspective, current computational limitations restrict us from applying such methods to a benchmark dataset of this scale. We expect that this will change in the coming years.

The work of Cunningham et al.¹⁷ and of Tal et al.⁴⁹ provided important initial benchmarks of vertex-corrected QSGW methods. Here, we present a systematic extension of these studies for a wider variety of materials. In doing so, we also validate previous results, a practice that, unfortunately, is still uncommon in modern computational materials science.

Throughout this study, we treat the QSG $\hat{W}$ band gaps as our reference standard, using them to benchmark other ab initio methods and to identify questionable experimental values.

Despite our original ambition to benchmark all 472 materials contained in the dataset, the steep computational scaling of GW calculations—formally $\mathcal{O}(N^4)$ for conventional implementations, where N is the number of atoms in the unit cell—makes materials with larger unit cells prohibitively expensive. Since the benchmark dataset contains materials up to $N = 28$, we are forced to introduce a system size limit. For method (i), we have performed calculations for materials with a maximum of 12 atoms per unit cell and atomic numbers up to 83 (excluding the lanthanides due to unavailable pseudopotentials). We have successfully converged and obtained band gaps for 286 (301) out of 332 systems using method (i), starting from an LDA (PBE) DFT calculation. The more demanding all-electron GW variants {(ii), (iii), (iv)} were further restricted to unit cells of six atoms or less, but no additional element restriction was imposed. There, we obtained converged band gaps for {156, 154, 101} of 171 materials.

The remaining compounds that failed to converge did so for technical reasons only. This mostly happened when we had insufficient memory to run the calculation or when the available CPU time was insufficient. High memory requirements often occurred for materials that required dense k -point grids to converge the band gap. In some cases, memory requirements were also high due to low unit cell symmetry, which indirectly increased the number of k -points used in a calculation. In other cases, it was due to the large number of electrons inherent in all-electron calculations and pseudopotentials containing many semicore states, e.g., Ag and Hg. In method (iv), the number of matrix elements of the Bethe–Salpeter Hamiltonian to be stored in memory scales as $(N_v \times N_c \times N_k)^2$, where N_v and N_c are the number of valence and conduction included in the BSE transition space, respectively, and N_k is the number of k -points. Therefore, if materials require either a dense k -point grid, such as Ge, or many bands, such as HgI, the memory requirements become enormous when vertex corrections in W are included in the self-consistency loop. This further limits the number of materials that we were able to calculate using method (iv). Despite these convergence and memory bottlenecks, we still ran many thousands of GW computations to obtain converged results, using several million CPU hours in total and up to several hundred gigabytes of memory per calculation, highlighting the steep scaling of these MBPT methods.

We have provided a spreadsheet containing the band gaps for all calculated materials as additional Supplementary Material, available online. For the complete dataset, including band structures and density of states (DOS) for all materials calculated using methods (ii–iv), refer to the Code and data availability statement.

Results

In the following, the four GW methods (i–iv) introduced above will be referred to as G_0W_0 -PPA, QPG_0W_0 , QSGW, and QSGW, respectively. All G_0W_0 -PPA gaps are converged to within 25 meV or better, while the more demanding full-frequency and self-consistent variants (ii–iv) are converged to within 100 meV or better, see Methods for details.

Before discussing the benchmark results, we first want to mention the fact that, depending on the DFT implementation used, the band gaps of the starting points of our GW calculation differed slightly from those reported in the original benchmark dataset^{3,4}. In some cases, materials classified as narrow-gap semiconductors in LDA calculations with a plane-wave basis set and projector augmented wave (PAW) pseudopotentials in the original benchmark dataset are metals in our norm-conserving pseudopotential and all-electron calculations. We analyze this in detail in Supplementary Note 1 of the Supplementary Information and attribute the discrepancies mainly to differences in the treatment of semicore states, which are included in norm-conserving (or all-electron) approaches but not in the PAW pseudopotentials selected by Borlido et al.^{3,4} based on the recommendations of the Materials Project^{52,53}, for elements such as Cu, Ag, Se, and Hg.

Benchmark of computational methods

To start, we benchmark all selected computational methods against experimental data using several statistical error metrics, which are summarized in Table 1. As is customary in benchmarks, we report the mean error (ME), the standard deviation of the error (σ), the mean absolute error (MAE), the root mean squared error (RMSE), and the mean absolute percentage error (MAPE). Here, the error is defined as $E_{\text{gap}}^{\text{calc.}} - E_{\text{gap}}^{\text{exp.}}$. We also investigated the effect of spin-orbit coupling (SOC) on the QPG_0W_0 and QSGW results (see Methods for details), referred to as QPG_0W_0 +SOC and QSGW+SOC, respectively, in Table 1. In all cases, SOC was applied *a posteriori* as a correction to the quasiparticle band structures. Histograms depicting the difference between computed and experimental band gaps for selected methods are presented in Fig. 1. For the sake of completeness, the histograms for the remaining methods are provided in Supplementary Note 2. Materials for which a method underestimates (overestimates) the experimental band gap are shown on the negative (positive) side of the histograms. We marked the ME and σ with solid and dashed orange lines, respectively.

Before analyzing the benchmark results, please note that the derived error metrics and histograms are for all materials for which a corresponding calculation is available. For example, Borlido et al.^{3,4} obtained band gaps for

Table 1 | Error metrics for all investigated calculation methods

	mBJ	HSE06	G_0W_0 @LDA-PPA	G_0W_0 @PBE-PPA	QPG_0W_0	QPG_0W_0 +SOC	QSGW	QSGW	QSGW+SOC
n	471	472	286	301	156	153	154	101	96
ME (eV)	-0.22	-0.10	-0.04	-0.12	0.08	-0.01	0.55	0.14	0.05
σ (eV)	0.68	0.85	0.75	0.82	0.54	0.55	0.63	0.45	0.47
MAE (eV)	0.50	0.53	0.53	0.60	0.37	0.37	0.64	0.31	0.30
RMSE (eV)	0.72	0.86	0.75	0.83	0.55	0.55	0.83	0.47	0.47
MAPE (%)	30	31	38	41	32	26	48	29	21
n	Materials for which experimental and computational data are available for all methods: 94								
ME (eV)	-0.42	-0.68	-0.14	-0.23	0.06	-0.03	0.56	0.13	0.04
σ (eV)	0.68	1.16	0.57	0.68	0.46	0.47	0.54	0.45	0.47
MAE (eV)	0.58	0.82	0.46	0.57	0.31	0.31	0.60	0.30	0.29
RMSE (eV)	0.80	1.34	0.58	0.71	0.47	0.47	0.78	0.47	0.46
MAPE (%)	35	38	35	37	29	20	42	30	21

The following metrics are given for the calculated band gaps using the methods described in the text: Dataset size (n), mean error (ME), standard deviation of the error (σ), mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The error is defined as $E_{\text{gap}}^{\text{calc.}} - E_{\text{gap}}^{\text{exp.}}$. The data for mBJ and HSE06 are taken from Borlido et al.^{3,4}. The metrics in the upper block are based on all available calculations for each method, as indicated by the n row. The lower block, on the other hand, restricts the comparison to the 94 materials for which data is available for all methods. Note that the MAPE has a tendency to preference methods that systematically underestimate band gaps, see explanation in the text. The best method for each metric is highlighted in bold.

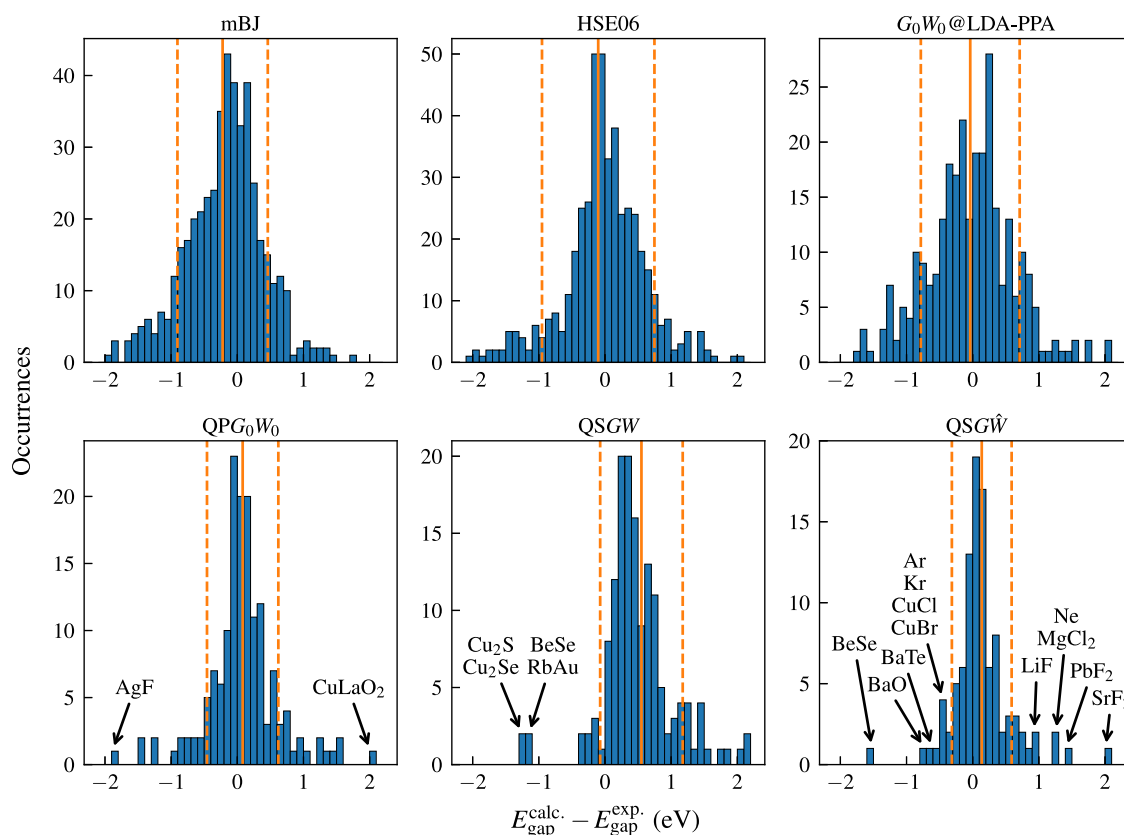


Fig. 1 | Histograms showing the differences between calculated and experimental band gaps. Each bin has a width of 0.1 eV. The mean error (ME) is indicated by a solid orange line, while the standard deviation (σ) is indicated by dashed orange

lines. Arrows in the histograms for the QPG₀W₀, QSGW, and QSGŴ calculations mark outliers, which will be analyzed in detail in the text.

all 472 materials, whereas we “only” performed QSGŴ calculations for 101 materials, as stated in Table 1. However, we also calculated all error metrics for a subset of 94 materials for which data is available for all investigated methods to facilitate a more direct comparison (cf. lower block of Table 1).

Regardless of the dataset used to evaluate the error metrics, the QSGW consistently emerges as the most accurate method, followed closely by the QPG₀W₀. Both mBJ and HSE06 tend to underestimate the experimental band gaps and exhibit high MAE and RMSE. G₀W₀@LDA-PPA calculations outperform G₀W₀@PBE-PPA calculations. Perhaps somewhat unexpectedly, the G₀W₀@LDA-PPA and G₀W₀@PBE-PPA calculations perform worse than mBJ in virtually all metrics when evaluated on the full datasets, except for the ME, where mBJ shows a systematic underestimation of the experimental band gap. While G₀W₀@LDA-PPA calculations perform better than HSE06 when evaluated on the full datasets, G₀W₀@PBE-PPA calculations do not. However, when the metrics of mBJ, HSE06, G₀W₀@LDA-PPA, and G₀W₀@PBE-PPA are compared on the subset of 94 materials for which data is available for all the investigated methods, the opposite is observed. Here, both G₀W₀-PPA methods outperform mBJ and HSE06, as the metrics for mBJ and HSE06 worsen dramatically, while those for the G₀W₀-PPA methods improve. No matter which dataset is chosen, the performance of mBJ and HSE06 is still impressive, considering that they are much less computationally demanding and easier to converge than GW calculations⁵⁴. The QSGW lags far behind the QPG₀W₀ and the QSGŴ due to its well-known tendency to systematically overestimate band gaps by about 15% (cf. ref. 17), and performs the worst overall in terms of error metrics. The inclusion of SOC in the QPG₀W₀ and QSGŴ calculations reduces the ME, bringing the calculated band gaps closer to the experimental values, modestly improving the ME and MAPE, but leaving other error metrics virtually unchanged.

At this point, we need to discuss and highlight an issue with the MAPE, a metric that is commonly used for band gap benchmarking. Firstly, the

MAPE for mBJ and HSE06 is much better than the MAPE of the G₀W₀-PPA methods and close to the MAPE of the QSGŴ. Secondly, the MAPE improves significantly when SOC is included in the QPG₀W₀ and the QSGŴ calculations. However, this is apparently contradictory to the other metrics and highlights a known limitation of MAPE: it systematically favors methods that underestimate band gaps⁵⁵. Furthermore, we note that the MAPE is highly affected by materials with small band gaps, where relatively small changes in the calculated band gap lead to extremely large percentage errors, see Supplementary Note 3. Therefore, the distribution of band gaps in a dataset (i.e., the presence of more or fewer materials with smaller or larger band gaps) affects this metric heavily. Consequently, we argue that MAPE should not be relied upon when evaluating the accuracy of computational methods for band gap prediction. We suggest using the MAE and the RMSE as primary metrics in the future, as is commonly done in the machine learning part of the materials science community.

We will now elaborate further on these findings and explain them in more detail using the histograms shown in Fig. 1, following the hierarchy of methodological complexity introduced earlier.

We begin by explaining why changing the starting point of a G₀W₀-PPA from LDA to PBE worsens all error metrics (see Table 1). At first, we were somewhat surprised by this, as our observations and experience suggest that G₀W₀@PBE-PPA calculations are used more frequently in practice than G₀W₀@LDA-PPA calculations. However, the superior performance of the LDA starting point compared to PBE for G₀W₀ calculations can be explained by the fact that G₀W₀ calculations rely heavily on fortuitous error cancellations, as discussed in detail in ref. 17. This is usually rationalized as follows: The band gap and the dielectric screening are inversely correlated⁵⁶. LDA underestimates band gaps by about 50%^{3,4} and thus overestimates the dielectric screening, while the random phase approximation (RPA)⁵⁷, used to calculate the screened Coulomb interaction W , systematically underestimates dielectric screening in semiconductors and insulators due to the

absence of excitonic effects¹⁷. In G_0W_0 @LDA-PPA calculations, the LDA-induced over-screening and the RPA-induced under-screening compensate in a fortuitous way, leading to band gaps in good agreement with experiment. Changing the starting point from LDA to PBE reduces this compensation because PBE predicts larger gaps^{3,4} and hence weaker screening than the LDA, leading to slightly worse G_0W_0 band gaps. This finding implies that the optimal G_0W_0 starting point is not simply “the most modern GGA” but the one whose self-consistent eigenvalues and screening happen to balance the systematic errors of the one-shot correction. That said, it remains plausible that starting from a DFT functional from the upper rungs of Perdew’s “Jacob’s ladder” (such as PBE0 or HSE06) could deliver better one-shot quasiparticle gaps, as hinted at in ref. 58. However, there is a clear trade-off: each rung above PBE increases the cost of the ground-state calculation considerably, and some functionals can be nearly as costly as the G_0W_0 step itself.

Compared to the G_0W_0 -PPA calculations, the QPG_0W_0 is in much better agreement with experiment and performs almost as well as the $QSG\hat{W}$. Since three parts of the QPG_0W_0 calculation method have changed compared to the G_0W_0 -PPA, we disentangle their individual impacts in the following: Firstly, we suspect that most of the improvements are due to the replacement of the PPA with a full-frequency integration of the dielectric screening. Secondly, since the starting point band gaps obtained by norm-conserving pseudopotential and all-electron calculations agree well, cf. Supplementary Note 1, the effect of the all-electron DFT starting point appears minor for most of the considered materials. Thirdly, the substitution of the linearized quasiparticle Eq. (1) for the effective potential given by Eq. (2) and following solution of the resulting QP equation most strongly affects materials where the LDA incorrectly ordered bands. Therefore, on average, this should also have a minor impact for most of the considered materials.

For the QSGW, we confirm the well-known fact that self-consistent RPA-based GW schemes that omit vertex corrections systematically overestimate band gaps—by about 15% in our benchmark. This is reflected in the much larger ME of 0.55 eV compared to the ME of 0.08 eV for the QPG_0W_0 . The origin of this systematic overestimation can be traced back to the error cancellation described above. The inclusion of self-consistency removes the dependence of the results on the starting point, thus correcting the initial band gap error of the DFT starting point. When W is then evaluated in the RPA, the dielectric screening is systematically underestimated due to missing excitonic effects, leading to band gaps that are systematically too large.

The $QSG\hat{W}$ alleviates this underestimation of the dielectric screening by including vertex corrections in W , i.e., excitonic effects. Therefore, the accuracy of $QSG\hat{W}$ largely stems from the cancellation of the systematic overestimation of QSGW, i.e., the $QSG\hat{W}$ relies on error cancellation. Looking at the corresponding histogram in Fig. 1 and the metrics in Table 1, we see that the $QSG\hat{W}$ does not improve the ME compared to the QPG_0W_0 , but it reduces σ considerably from 0.54 eV to 0.45 eV. However, when comparing the $QSG\hat{W}$ and QPG_0W_0 on the subset of 94 materials for which data is available for all the investigated methods, we observe that both perform about equally well with respect to all error metrics except the ME.

While both the QPG_0W_0 and the $QSG\hat{W}$ method perform exceptionally well, they are hindered in practice by well-known limitations. The QPG_0W_0 omits self-consistency and is therefore sensitive to the DFT starting point. The $QSG\hat{W}$ is computationally demanding as it requires the evaluation of vertex corrections. A common, pragmatic alternative that retains the benefits of self-consistency while alleviating the systematic RPA underscreening without calculating expensive vertex corrections is to rescale the quasiparticle QSGW self-energy by mixing it with a fraction of the LDA DFT exchange-correlation potential.

$$\Sigma_0^{\text{mix}} = (1 - \alpha)V_{xc}^{\text{LDA}} + \alpha\Sigma_0^{\text{QSGW}} \quad (4)$$

This empirical approach closely resembles the construction of a hybrid functional, such as PBE0⁵⁹. Previous work has often adopted the pragmatic

choice of $\alpha = 0.8$ ^{60,61}, while Chantis et al.⁶² fitted α individually for each material to best reproduce the experimental band gap at 0 K. Here, we systematically tested $\alpha = 0.8$, which we denote as αQSGW , for the 154 materials where converged QSGW results were available. The results are analyzed in detail in Supplementary Note 4, where they are compared against both the $QSG\hat{W}$ and experimental band gaps. Overall, the results show that the αQSGW consistently mitigates the systematic overestimation of band gaps inherent to the QSGW, achieving band gaps and errors that are nearly identical to those of the $QSG\hat{W}$. Consequently, the αQSGW reproduces experimental values with an accuracy nearly identical to that of the $QSG\hat{W}$, while being substantially less computationally demanding. Thus, it is an attractive alternative for large-scale or high-throughput studies. In addition to QPG_0W_0 and αQSGW calculations starting from an LDA calculation, another practical route for such studies may be to perform QPG_0W_0 calculations starting from more advanced functionals, such as mBJ or HSE06.

The band structures of the exemplary materials MgO and ScN, shown in Fig. 2, illustrate the points made above. For MgO (left panel), the $QSG\hat{W}$ (black lines) and QPG_0W_0 (red lines) are nearly identical. For both methods, the band gap agrees well with the experimental value. The systematic band gap overestimation of the QSGW is evident in all conduction bands. For ScN (right panel), however, the QPG_0W_0 (red lines) deviates from the $QSG\hat{W}$ (black lines) by about the same amount as the QSGW, but in the opposite direction. The $QSG\hat{W}$ band gap agrees well with experimental data. This demonstrates that, while the QPG_0W_0 relies on a fortuitous cancellation of errors that can produce electronic structures in line with the much more expensive QSGW on average (cf. Table 1), its accuracy for specific materials may vary. Note also that the LDA band structure (gray lines) for ScN is that of a (semi-)metal.

The inclusion of SOC as an *a posteriori* correction to the QPG_0W_0 and $QSG\hat{W}$ results significantly improves the calculated band gap of materials containing heavy elements from the lower parts of the periodic table. This is consistent with the fact that the spin-orbit coupling constant scales approximately with Z^4 in the atomic limit. A prime example is PbTe, where the inclusion of SOC reduces the $QSG\hat{W}$ band gap from 0.89 eV to 0.34 eV, bringing it much closer to the experimental value of 0.19 eV⁶³. Similar improvements are observed for other heavy element compounds such as SnTe, CdTe, and PbSe, cf. Supplementary Material. For a systematic overview, Supplementary Note 5 provides the SOC corrections for all materials as a function of the maximum atomic number Z_{max} in each material, and illustrates how a practical Z-cutoff criterion could be applied in future large-scale benchmarks. These results highlight the importance of including SOC when aiming for high accuracy in heavy-element materials.

Investigating outliers

Since QPG_0W_0 and $QSG\hat{W}$ calculations proved to be highly accurate, we examined materials with significant deviations from the reported experimental band gaps in detail. Additionally, we investigated materials for which the QSGW significantly underestimates the experimental band gap. In other words, we examined cases that contradict the tendency of the QSGW to overestimate band gaps. Identified outliers are highlighted in Fig. 1 with arrows in the associated histograms. These are often “unusual” compounds, such as an alkali auride. Some outliers are caused by the absence of electron-phonon interactions in the calculations. In other cases, strong electron correlations are not adequately described, even by the most advanced GW variants employed in this study. More often, however, experiments have likely been misinterpreted or were based on suboptimal samples.

We would like to preface the following section with a disclaimer: We are aware that the analysis below is quite detailed and may sometimes feel laborious to read. However, we believe that thoroughly documenting these cases is valuable because it provides important context for interpreting the benchmark data and establishes a basis for revising experimental band gaps in the future. Additionally, please note that we are not experts on any of the discussed materials. On the contrary, we only became aware of them due to their above-average error metrics. In each case, we can only provide a brief

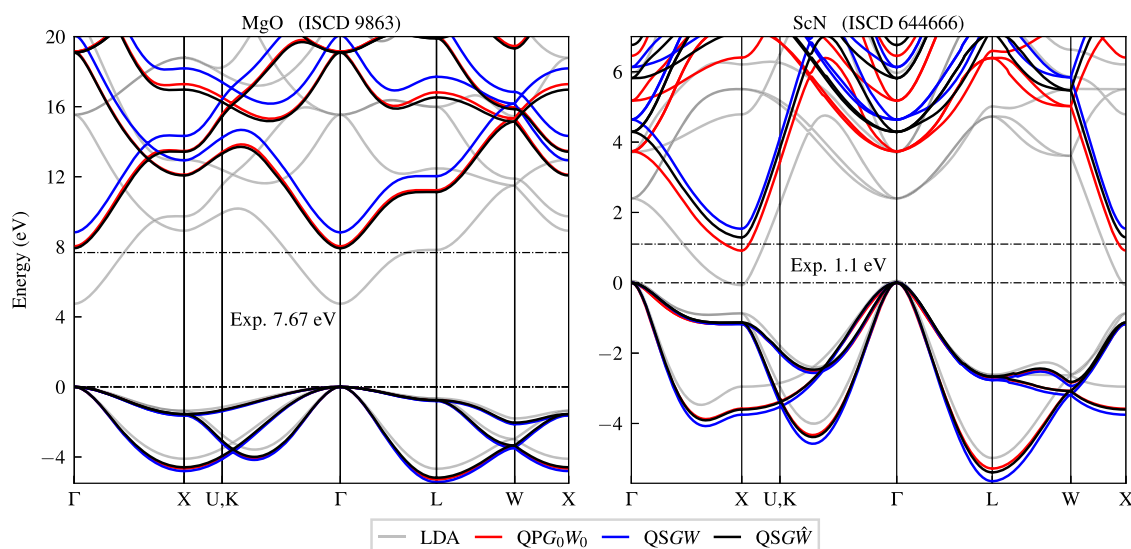


Fig. 2 | Band structures calculated using different methods for the exemplary materials MgO (left panel) and ScN (right panel). The dashed horizontal lines mark the experimental band gaps. The band gap for MgO is taken from ref. 63, and the band gap for ScN is taken from Grümbel et al.⁹⁵ as discussed at the end of the

Results section. The structures used for the calculations are identified by their respective ICSD database identifiers^{29–31}, which are given in the panel titles. The symmetry path was taken from ref. 109.

Table 2 | Band gaps (eV) from LDA, QSGW, QSGW-hat, and experiment for the QSGW outliers in our benchmark set

Material	LDA (eV)	QSGW (eV)	QSGW-hat (eV)	Exp. (eV)	Exp. Ref.
CuLaO ₂	2.48	5.09	4.36 ^a	2.34 ^b	65
Cu ₂ S	0.00	0.00	—	1.21 ^c	63
Cu ₂ Se	0.00	0.00	—	1.20 ^c	63
RbAu	0.12	1.61	—	2.75 ^d	74

Conjectures for why these materials appear as outliers, i.e., why the deviations between experiment and theory are particularly large, are summarized in the footnotes annotated by superscripts. A detailed discussion of each material is provided in the main text. A dash indicates that the band gap was not computed using this method.

^aSee Supplementary Note 6.

^bOptical vs. quasiparticle gap; strong excitonic/phonon effects.

^cStructural disorder; missing short-range correlations.

^dPossible correlation effects in aurides; limited literature.

overview of the literature and add our, hopefully sometimes helpful, perspective as computational materials scientists. This may inspire new investigations in theoretical and experimental solid state physics, which some of the unusual materials certainly deserve.

For the QPG₀W₀ calculations (lower left panel of Fig. 1), we looked at the outliers AgF and CuLaO₂. The value of 0.91 eV for the QPG₀W₀ band gap of AgF severely underestimates the experimental band gap of 2.8 ± 0.3 eV⁶⁴. In contrast, the QSGW band gap of 3.18 eV agrees well with the experiment. We attribute this behavior to the metallic ground state predicted by the LDA for AgF. Many-body corrections are initially required to open a gap, and self-consistency must be achieved to make that gap substantial.

CuLaO₂ is a delafossite oxide that has been studied for its use in photocatalytic hydrogen production⁶⁵. It should not be confused with La₂CuO₄, which is well-known for its high-temperature superconductivity and unique electronic properties, cf. ref. 17. The CuLaO₂ QPG₀W₀ band gap of 4.35 eV significantly overestimates the experimental band gap of 2.34 eV⁶⁵. Advancing to QSGW enlarges the band gap to 5.09 eV, which increases the error and pushes the point off-scale in the QSGW histogram in Fig. 1. Because a QSGW-hat using our benchmark workflow (cf. Methods) exceeded our memory and compute budget, we recomputed the QSGW and

QSGW-hat using a reduced basis set and a truncated BSE transition space, see Supplementary Note 6. These QSGW and QSGW-hat calculations yielded band gaps of 4.93 eV and 4.36 eV, respectively. Although these values are not fully converged, we note that the observed band gap renormalization caused by the inclusion of vertex corrections in W of 0.57 eV is relatively large. Based on a limited study of the influence of the BSE transition space on the QSGW-hat band gap, we estimate that the converged QSGW-hat renormalization is even greater, see Supplementary Note 6. However, the resulting band gap still exceeds the experimental value by about 2 eV. This exactly mirrors the behavior of the related delafossite oxide, CuAlO₂, analyzed by Cunningham et al.¹⁷. There, the authors demonstrate that the almost dispersionless Cu 3d valence band causes a deeply bound exciton, lowering the optical gap to approximately 3.2 eV, whereas they obtained a fundamental QSGW-hat gap of 4 eV. The authors further argue that including electron-phonon processes may lower the optical gap further, as their QSGW-hat+BSE optical gap is still larger than the experimental reports for CuAlO₂¹⁷. Additionally, Cunningham et al.¹⁷ noted that the optical absorption edge of CuAlO₂ is strongly dependent on preparation and post-annealing conditions, cf. ref. 66. Examining the element-resolved QSGW band structure in Supplementary Note 6 reveals a similar, almost dispersionless valence band composed of Cu states, with minor contributions from O states, for CuLaO₂. Since the experimental band gap of CuLaO₂ used in the benchmark comes from a reflectivity measurement, i.e., an optical measurement, we attribute the observed discrepancy to the fact that the measurement most likely reports the optical gap rather than the fundamental electronic band gap. We also believe that electron-phonon processes and influences from the sample growth process, similar to those in CuAlO₂, may influence the optical gap of CuLaO₂. Therefore, further experiments on CuLaO₂, as well as a full-scale, in-depth QSGW-hat+BSE study, may be warranted to determine its true optical/fundamental gap.

Table 2 provides a summary of the outlier materials, including CuLaO₂ discussed above, as well as the QSGW outliers that are analyzed in detail below. We begin with the antiferrofluorites Cu₂S and Cu₂Se, both of which are found to be semimetallic within the QSGW. In the case of Cu₂S we reproduce the results of Lukashev et al.⁶⁷, i.e., that the ideal antiferrofluorite Cu₂S remains semimetallic in both LDA and QSGW because of the triple-degenerate Cu 3d valence band maximum at the Γ point that overlaps the Cu 4s conduction band. Lukashev et al.⁶⁷ showed that symmetry breaking through small deviations from the ideal antiferrofluorite positions opens a

Table 3 | Band gaps (eV) from LDA, QSGW, QSG $\hat{W}$, and experiment for the QSGW outliers in our benchmark set

Material	LDA (eV)	QSGW (eV)	QSG $\hat{W}$ (eV)	Exp. (eV)	Exp. Ref.
BaTe	1.40	3.12	2.79	3.40 ^a	76
BeSe	2.47	4.35	3.98	5.50 ^a	77
BaO	1.83	4.54	4.07	4.80 ^b	80
CuCl	0.39	3.64	2.84	3.40 ^c	63
CuBr	0.29	3.26	2.60	3.07 ^c	63
Ar	8.18	14.57	13.70	14.15 ^c	110
Kr	6.77	12.00	11.15	11.59 ^c	110
LiF	8.84	15.84	14.53	13.60 ^d	111
Ne	11.41	23.33	22.74	21.48 ^d	110
MgCl ₂	5.24	9.61	8.79	7.50 ^e	88
PbF ₂	4.25	7.73	7.09	5.68 ^{e,f}	63
SrF ₂	6.61	12.21	11.61	9.60 ^e	89
ScN	0.00	1.54	1.29	0.90 ^b	93

Conjectures for why these materials appear as outliers, i.e., why the deviations between experiment and theory are particularly large, are summarized in the footnotes annotated by superscripts. A detailed discussion of each material is provided in the main text.

^aExperiment likely reports the direct band gap.

^bConflicting experimental reports.

^cMissing vertex corrections in Σ likely important.

^dOverestimation caused by missing EPR.

^eExperiment likely reports the optical gap.

^fUnknown cause.

narrow gap. However, severe structural disorder, non-stoichiometry, and other effects, as discussed in ref. 67, complicate optical measurements, and make calculating the “experimental” gap of 1.1–1.2 eV difficult for Cu₂S⁶³.

We conjecture that similar physics is at play in Cu₂Se. References 68–70 indicate that strong electronic correlations may also play an important role, as expected for partially filled, degenerate *d* bands. Zhang et al.⁶⁹ showed that the treatment of Cu₂Se with mBJ + *U*, using a moderate *U* = 4 eV for the Cu *d* states, opens a small band gap of 0.5 eV, still underestimating the experimental band gap of 1.2 eV⁶³. This result is also consistent with the PBE0 + *U* calculations of Räsander et al.⁶⁸, although they used an unusually large value of *U* = 10 eV. A band gap of 0.5 eV was also obtained though an HSE06 + *U* calculation with *U* = 4 eV by Klan et al.⁷⁰. Because the RPA screening used in the QSGW is a poor approximation of short-range correlations⁷¹, and DFT + *U* calculations effectively model short-range, on-site correlations, we presume that short-range correlations are important for obtaining an accurate electronic structure of Cu₂Se. Thus, a more complete picture of the electronic structure of Cu₂Se will likely require treating electronic correlations through methods such as QSGW + DMFT⁷², as well as the inclusion of substantial structural disorder and non-stoichiometry mentioned in refs. 67,69,70. We further conjecture that approaches coupling advanced many-body methods with realistic structural models may be required to obtain band gaps that align with experimental results for Cu₂S, Cu₂Se, and possibly Cu₂Te.

In the case of RbAu, a representative of the small and special class of insulating aurides in which Au forms a simple isolable monoatomic anion⁷³, we are unsure what exactly causes the deviation of the calculated QSGW band gap of 1.61 eV to the experimental band gap of 2.75 eV⁷⁴. Interestingly, ref. 74 states that the band gap of RbAu reduced to 1.35 ± 0.1 eV upon melting at 510 °C, which appears to be in better agreement with our calculations. However, we did not calculate the liquid state of RbAu. Earlier work by Liu⁷⁵ describes RbAu as “more like a metallic alloy, with optical properties similar to pure gold in the visible region”, referring to a reflectivity feature near 2 eV. Based on this limited literature for RbAu, we encourage the experimental community to remeasure the optical response of RbAu under well-controlled conditions and encourage theoretical physicists to look closer into the issue of electronic correlations in aurides.

Finally, we examine the outliers of our QSG $\hat{W}$ calculations (lower right panel in Fig. 1). The fact that these are apparently numerous is ultimately also a downside of the success of this method, namely its low standard deviation. Given the exceptional accuracy and consistency of the QSG $\hat{W}$ method across the benchmark dataset, significant deviations typically suggest either experimental ambiguities or the presence of physics beyond what is captured by the current (purely electronic) many-body treatment. A condensed summary of the discussion on the individual materials is provided in Table 3.

Among the materials for which QSG $\hat{W}$ underestimates the experimental band gaps, the most pronounced outliers are BaTe, BeSe (also highlighted in the QSGW histogram), and BaO. For BaTe, Martin et al.⁷⁶ measured a band gap of 3.4 eV using photoelectric emission. Analyzing the QSG $\hat{W}$ band structure for BaTe, we found an indirect band gap of 2.79 eV along the Γ -X symmetry line and a direct band gap of 3.25 eV at the X point. Therefore, we argue that Martin et al.⁷⁶ measured the direct band gap at the X point rather than the fundamental indirect gap, explaining the apparent underestimation highlighted in the histogram. A similar case can be made for the indirect insulator BeSe, where Wilmers et al.⁷⁷ report an optical absorption onset E_0 = 5.5 eV derived from ellipsometry data, whereas our QSG $\hat{W}$ calculation predicts an indirect gap of 3.98 eV along the Γ -X symmetry line and a direct gap of 6.07 eV at the Γ point. We therefore argue that Wilmers et al.⁷⁷ reported the direct optical gap at the Γ point for BeSe. A later resonant inelastic X-ray scattering study by Eich et al.⁷⁸ measured a direct band gap of 5.7 ± 0.2 eV at the Γ point and an indirect band gap of 3.8 ± 0.2 eV for BeSe. These values agree much better with our calculated band gaps. The discrepancy between the values for the direct gap reported by Wilmers et al.⁷⁷ and Eich et al.⁷⁸ likely arises from the fact that Wilmers et al.⁷⁷ measured the optical gap rather than the fundamental electronic band gap. The main difference between both is the exciton binding energy, estimated here to be around 80 meV⁷⁹. The remaining difference is probably due to other experimental uncertainties.

For BaO, our QSG $\hat{W}$ calculation yields a band gap of 4.07 eV, whereas the experiment by McLeod et al.⁸⁰ reports a band gap of 4.8 eV. However, two older optical measurements report much smaller onset energies for the optical absorption in BaO: 4.10 eV at room temperature by Saum et al.⁸¹ and 4.043 eV for the first bound exciton at 2 K by Kaneko et al.⁸². When estimating the exciton binding energy of BaO to be around 80 meV⁷⁹, our QSG $\hat{W}$ band gap of 4.07 eV aligns exceptionally well with these older experiments. Moreover, McLeod et al.⁸⁰ note a secondary emission band arising from BaCO₃ surface carbonation in their spectra, indicating residual CO₂ contamination that could have biased their surface-sensitive X-ray emission/absorption spectroscopy band gap determination. Taken together, these observations call into question the accuracy of the 4.8 eV band gap reported in ref. 80 for BaO. We presume that throughout the dataset, it is not inconceivable that incorrect band gaps may be reported for other materials, especially for those we did not calculate with the QSG $\hat{W}$ method.

The remaining cases in which our QSG $\hat{W}$ calculations underestimate the experimental band gap stem from well-documented shortcomings inherent to the method. These have been discussed in detail by Kutepov¹⁵ and by Cunningham et al.¹⁷. We provide only a brief and limited summary here and refer the reader to these publications for a more thorough discussion.

The quasiparticle approximation employed by the QSG $\hat{W}$ approach neglects full self-consistency in Green’s function calculations. Kutepov¹⁵ analyzed this in detail by starting out from the approximations invoked in Questaal and systematically removing them. The final conclusion was that additional approximations not inherently assumed in Hedin’s equations are required to achieve “good results” when employing the quasiparticle approximation. Specifically, these approximations include the TDA, the static approximation of the BSE kernel, and excluding vertex corrections in Σ ^{15,17}. There, it is also noted that omitting vertex corrections in Σ helps to avoid the destructive effect of *Z*-factor cancellation (see Appendix A of ref. 10). However, this cancellation argument is only strictly valid in the long-range limit and should not be used as a general justification⁴⁹. Recent

studies on molecules and small silicon clusters have shown that vertex corrections in W and Σ nearly cancel each other, both for G_0W_0 ⁸³ and for QSGW⁸⁴ calculations. Thus, when the QSGW band gap underestimates experimental results, it seems plausible that vertex corrections in Σ become essential. Simply put, when the QSGW works well, QSGW will be a poor approximation. Therefore, as previously mentioned, the apparent success of QSGW arises because the systematic overestimation of band gaps in QSGW is fortuitously canceled. In a somewhat ironic way, the accuracy of the QSGW seems to rely on fortuitous error cancellation, somewhat similar to G_0W_0 calculations.

A notable example of these shortcomings of the QSGW method is CuCl^{15,17}, where the inclusion of vertex corrections in W within the quasi-particle self-consistency severely underestimates the band gap. This behavior is also clearly observed in our benchmark, see Fig. 1. Our QSGW band gap for CuCl of 2.84 eV agrees with the values reported by Kutepov¹⁵ and Cunningham et al.¹⁷. Furthermore, in agreement with refs. 15,17, we find that our QSGW band gaps of 3.64 eV for CuCl and 3.26 eV for CuBr are closest to the experimental values of 3.4 eV and 3.07 eV⁶³, respectively. This nicely underlines the point above: when the QSGW works well, e.g., for CuCl and CuBr, the additional W -only vertex correction in the QSGW method actually worsens the results. For rare-gas solids, Ar and Kr, for which experiment lies between the QSGW and the QSGW, only partial cancellation appears to occur between the vertex corrections in W and Σ . Cunningham et al.¹⁷ point out that such underestimations tend to occur when the highest occupied bands are flat and almost dispersionless, matching our observations for CuCl, CuBr, and the rare gas solids, Ar and Kr.

We turn now to the cases where QSGW overestimates the experiment, focusing first on LiF and solid Ne. Like all other methods evaluated in this benchmark, our QSGW results omit electron-phonon interactions. Electron-phonon renormalization (EPR) typically reduces band gaps when included in calculations^{15,17}. For light-element, wide-gap insulators, this omission leads to significant overestimations, as is the case for LiF and Ne. What distinguishes the QSGW from the other methods is its exceptional accuracy and low standard deviation across the dataset, which allows such missing contributions to manifest most clearly as large deviations from the experiment. In other words, these discrepancies primarily highlight cases in which EPR, and potentially other electronic correlations that are missing from the QSGW, become relevant in order to achieve quantitative agreement with the experiment. Consequently, the QSGW band gaps of these materials are noticeable outliers.

Interestingly, in some materials, the omission of EPR can be offset by the systematic underestimation of the band gap introduced by the W -only vertex correction. Diamond is the canonical example: the QSGW band gap of 5.64 eV agrees with the measured value of 5.5 eV⁶³ only because the missing EPR contribution of about 0.4 eV^{85–87} nearly cancels the underestimation from the vertex correction in W . This observation has also been made previously by Kutepov¹⁵. Once again, we emphasize that the accuracy of the QSGW method appears to depend on fortuitous error cancellation, somewhat similar to G_0W_0 calculations.

In contrast, the discrepancies between the published experimental values for MgCl₂⁸⁸, PbF₂⁶³, and SrF₂⁸⁹ and our QSGW calculations are presumably due to the fact that the reported values correspond to optical gaps (i.e., optical absorption onsets) rather than fundamental electronic band gaps. This interpretation is supported by other reported measurements for MgCl₂ and SrF₂: For MgCl₂, Kinno et al.⁹⁰ reported from reflection spectra an $n = 1$ exciton at 8.5 eV and estimated a band gap of 9.2 eV. This value is more consistent, though still not perfectly aligned, with our QSGW band gap of 8.79 eV than with the significantly smaller (presumably optical) band gap of 7.5 eV reported by Sugiura⁸⁸. For SrF₂, Rubloff⁹¹ measured an exciton at 10.6 eV and a band gap of 11.25 eV from reflectance spectra, which aligns well with our QSGW value of 11.61 eV. We again presume that optical gaps have been reported for other materials in the benchmark dataset, though such cases are often difficult to identify without reexamining the original experiments or carrying out a QSGW calculation for comparison. However, the situation remains uncertain for PbF₂. Both Findley et al.⁹²

and Madelung⁶³ report similar band gaps of 5.84 eV and 5.68 eV, respectively, which are much smaller than our QSGW band gap of 7.09 eV. Furthermore, the measured exciton binding energy of only 0.24 eV at 300 K reported by Findley et al.⁹² is unusually small for a highly ionic fluoride with weak dielectric screening⁷⁹. Therefore, we believe that a more detailed theoretical study of PbF₂ is warranted.

Since the aforementioned misreported band gaps (direct vs. indirect and optical vs. fundamental) and materials, such as Cu₂S and Cu₂Se, for which the experimental crystal structures probably differ from the high-symmetry structure used in the calculations, bias the error statistics, we have recalculated all metrics for a revised dataset, see Supplementary Note 7. In addition to excluding the eight previously mentioned materials, i.e., CuLaO₂, Cu₂S, Cu₂Se, MgCl₂, PbF₂, SrF₂, RbAu, and BaTe, we also adjusted the reference experimental band gaps for BeSe and BaO (see above), as well as ScN.

Regarding ScN, we observed that Cunningham et al.¹⁷ also encountered difficulty in explaining the discrepancy between their QSGW of 1.27 eV (here: 1.29 eV) and the experimental values of 0.9 eV by Al-Brithen et al.⁹³ and 0.92 eV by Deng et al.⁹⁴, even when approximating the band gap reduction caused by EPR to be around 0.1 eV. However, a more recent work by Grömbel et al.⁹⁵, presumably using a cleaner sample, reported a band gap of 1.1 ± 0.1 eV, aligning more closely with the results of Cunningham et al.¹⁷ and this study.

Furthermore, while analyzing the performance of α QSGW, we identified LiCoO₂ as an additional outlier and also excluded it from the dataset. This observation is consistent with the conclusions by Radha et al.⁹⁶, who showed that the experimentally reported value corresponds to the optical rather than the fundamental electronic band gap. For details and discussion of possible additional, yet unconfirmed, outliers, see Supplementary Note 4. We also reassessed the performance of α QSGW on this revised dataset, see Supplementary Note 7.

To make the results comparable to those of previous publications that also used the Borlido et al.^{3,4} benchmark dataset, the main text still reports results on the original dataset. However, we recommend using the revised benchmark dataset (provided as additional Supplementary Material available online), which omits or reclassifies materials with incorrectly reported band gaps. We also strongly believe that the original benchmark dataset contains more materials with misreported band gaps and structures, particularly for materials with larger unit cells, for which we did not perform any GW calculation.

Discussion

We conducted a systematic benchmark for band gaps of solids, comparing MBPT methods, specifically, various GW approximations, with two state-of-the-art DFT functionals, namely mBJ and HSE06, on a large dataset of non-magnetic semiconductors and insulators. Addressing the central research question raised in this study—“How well do MBPT calculations perform relative to the best available DFT functionals when predicting the band gaps of solids?”—our findings clearly demonstrate the superior performance of advanced GW schemes. Specifically, we found that GW methods such as the QPGW₀ and the QSGW significantly outperform the current best DFT functional (mBJ and HSE06) and simpler GW variants (G_0W_0 -PPA). As expected, the QSGW emerged as the most accurate method overall.

Interestingly, although the computationally cheaper G_0W_0 approaches exhibited good average accuracy, their reliability heavily depends on fortuitous error cancellations. These cancellations arise due to balancing opposite systematic errors, i.e., overestimated dielectric screening from the DFT starting point and underestimated dielectric screening due to the use of the RPA and the absence of excitonic effects (vertex corrections in W). Our benchmark again highlights the sensitivity of G_0W_0 results to the choice of the DFT functional used as a starting point. Notably, contrary to common practice, performing a G_0W_0 calculation starting from an LDA DFT calculation performs better on average than those starting from a PBE DFT calculation.

Furthermore, the observed performance of G_0W_0 -PPA calculations compared to DFT calculations depends heavily on whether the full dataset or a subset of materials with complete data coverage is considered. We argue that the complete data coverage subset provides a fairer basis for comparison, and on this basis, G_0W_0 -PPA slightly outperforms the meta-GGA functional mBJ and notably surpasses the hybrid functional HSE06. Nonetheless, both DFT functionals remain substantially more computationally efficient and easier to converge (cf. ref. 54). Particularly, the excellent performance combined with computational affordability explains the widespread adoption of HSE06, especially for large-scale applications involving, e.g., surfaces⁹⁷ and defects⁹⁸. However, caution should be exercised when using the mBJ functional, given its semi-empirical nature, as it was specifically fitted to experimental band gaps³². Moreover, mBJ and other meta-GGA functionals often exhibit stability issues, convergence difficulties, and generally struggle to provide accurate electronic densities, exchange-correlation energies, and band gaps simultaneously⁹⁹.

Even though the DFT functionals used for comparison in this benchmark are surprisingly good and well-performing, our findings show that MBPT in the form of different GW variants can contribute a lot to our understanding, and that ultimately, at least the QSGW method simply has the superior predictive power. Note that the QPG_0W_0 and $\alpha QSGW$ are attractive alternatives to the QSGW due to their intrinsic error compensation and the lower costs. While more expensive than the QPG_0W_0 , the $\alpha QSGW$ retains the benefits of self-consistency. However, starting QPG_0W_0 calculations from more advanced functionals, such as mBJ or HSE06, may improve accuracy and reduce its dependency on the starting point. Naturally, this raises the question of whether the additional cost of such starting points remains competitive with an $\alpha QSGW$ calculation starting from an LDA calculation. We believe that future work should address this matter.

Despite recent methodological advances, limitations still remain in current MBPT approaches. For instance, despite its superior accuracy, the QSGW approach systematically underestimates the band gaps of materials with flat or almost dispersionless highest occupied bands^{15,17}. Nevertheless, the accuracy of the QSGW method is arguably already within the experimental tolerance for most materials, which underscores its strength as a high-fidelity predictive tool. Our detailed examination of outliers also revealed several experimental ambiguities, such as confusion between optical and fundamental band gaps, as well as between direct and indirect transitions, and the influence of structural anomalies. Additionally, we emphasize that all methods evaluated in our benchmark omit EPR. Correcting for these effects on the experimental side would require carefully annotating the Borlido dataset³⁴ with measurement temperatures and reference band gaps corrected for zero-point and finite-temperature renormalization effects, as well as further cleaning and standardizing the entries. In particular, distinguishing between optical and fundamental electronic band gaps, or ideally providing both when available, would substantially improve the interpretability of such benchmarks and could allow previously excluded materials to be reinstated in future versions of the dataset. Establishing such a curated, temperature-consistent reference dataset would be an important milestone for the community, enabling more accurate, meaningful comparisons between theory and experiment. On the theoretical side, systematic GW +EPR calculations have been successfully demonstrated, but thus far only for test systems¹⁰⁰ and not in large-scale benchmarks. However, we believe that this situation will change in the coming years as workflows for combining GW methods with electron-phonon corrections in solids are already published¹⁰¹, making such approaches increasingly feasible for large-scale benchmarks.

Finally, our benchmark also provides critical guidance for constructing reliable datasets for machine learning applications in materials science from a broader perspective. Despite the high computational cost of MBPT calculations, the strategic use of MBPT methods to generate small, high-quality datasets for transfer learning seems promising. This approach could significantly improve the reliability and accuracy of future machine-learning-assisted materials predictions while balancing computational affordability with the necessity of precision in materials discovery.

Methods

All calculations described below use the same experimental geometries from the ICSD^{29–31} that were used in the DFT benchmark by Borlido et al.³⁴. The corresponding material identifiers can be found either in the DFT benchmark dataset by Borlido et al.³⁴ or in the provided Supplementary Material. All structures were reduced to their primitive standard structure using Spglib¹⁰². For all k-point grids (i.e., $k_x \times k_y \times k_z$) used in calculations described below, each subdivision along the x , y , and z directions (i.e., k_x , k_y , and k_z) was rounded up to the nearest even integer. We provide a great deal of detail in the Methods section because the value of any benchmark depends on its reproducibility and its character as “good practice” template for own work.

G_0W_0 -PPA using Quantum ESPRESSO and Yambo

We performed DFT calculations with the plane-wave code Quantum ESPRESSO (version 7.1)^{38,39} as starting point for the G_0W_0 -PPA calculations, using LDA and PBE as exchange-correlation functionals³⁷. For the LDA calculations, we used optimized norm-conserving Vanderbilt (ONCV) pseudopotentials from the PseudoDojo project (version 0.4.1)^{103,104}. For the PBE calculations, we used ONCV pseudopotentials from the SG15 library (version 1.2)¹⁰³.

The G_0W_0 -PPA calculations were carried out using the Yambo code (version 5.2.4). The frequency dependence of the dynamical screening W was approximated through the Godby-Needs plasmon-pole approximation³⁶. To accelerate the convergence of Σ with respect to the number of empty bands, N_b , the Bruneval-Gonze technique was employed¹⁰⁵. The $\mathbf{q} \rightarrow 0$ divergence of the Coulomb potential was treated with the random integration method⁴⁰ as implemented in Yambo. The same plane-wave cutoff energy was used in Yambo to expand the KS wave functions and densities in plane waves as was used for the converged DFT calculations in Quantum ESPRESSO.

For the G_0W_0 @LDA-PPA and G_0W_0 @PBE-PPA calculations, we used the following workflow:

(i) As a first step, we converged the DFT plane-wave cutoff energy and k-point grid. The starting value for the plane-wave cutoff energy was set to the value suggested by the respective pseudopotentials, and for the k-point grid, the initial value was set to a structure-independent reciprocal density of $n_k = 1500$ per atom as defined in pymatgen¹⁰⁶. We then iteratively increase the plane-wave cutoff energy by 5 Ry, while keeping n_k fixed, until the change in the total energy per atom is smaller than 1 meV. Then, using the found plane-wave cutoff energy, we converged the total energy change per atom with respect to the k-point grid to the same 1 meV threshold. For this, we increased the k-point grid, $k_x \times k_y \times k_z$, in the following manner: We increased the k-point grid density, n_k , until the number of subdivisions in some direction of the resulting new k-point grid was greater than that of the previous k-point grid, i.e., $k_i^{n+1} > k_i^n$ for some $i = \{x, y, z\}$. Then, we performed a DFT calculation using the new, denser k-point grid and compared the total energy per atom to the previous one. This process was repeated until the total energy change per atom converged below 1 meV.

(ii) Similar to Borlido et al.³⁴, we converge the band gap with respect to the k-point grid by increasing the k-point grid from the previously found n_k until the changes in the band gap are below a threshold of 25 meV. Here, the plane-wave cutoff energy is set to the converged value from step (i).

(iii) Starting from converged KS wavefunctions and eigenvalues from step (i), we converged the G_0W_0 -PPA corrections with respect to the number of empty bands, N_b , the plane-wave cutoff energy used for the dielectric matrix, G_{cut} , and the GW k-point grid density n_k^{GW} . These three parameters were converged using a workflow previously introduced by us⁵⁴. The starting points and step sizes of the parameters were set to the values from ref. 54. The convergence threshold for the direct band gap at the Γ point was set to 25 meV.

(iv) To ensure the highest possible accuracy at a moderate increase in computational cost for the G_0W_0 -PPA calculations, we increased the parameters found in (iii) (i.e., N_b , G_{cut} , and n_k^{GW}) by one additional step and performed a final G_0W_0 -PPA calculation. To save computational resources,

we only calculate the G_0W_0 -PPA band gap correction at the Γ point for the valence band maximum and the conduction band minimum.

(v) The band gap correction obtained from the G_0W_0 -PPA calculation at the Γ point was then used as a “scissor operator”, i.e., the obtained value was added to the converged DFT gaps from step (ii), resulting in the band gaps reported in this study.

All-electron GW calculations in Questaal

We performed all-electron DFT, QPG $_0W_0$, QSGW, and QSG $\hat{W}$ calculations in the LMTO code Questaal (version 7.14.1). For details regarding the all-electron LMTO implementation, QSGW methodology, and how vertex corrections are included in W , we refer to the work by Pashov et al.⁴², Kotani et al.¹⁰, and Cunningham et al.¹⁷, respectively.

A particularly useful feature of the Questaal code is its ability to interpolate Σ_0 , see Eq. (2), onto an arbitrary k -point grid⁴². To take advantage of this feature, GW calculations in Questaal typically use a different k -point grid for calculating Σ than for solving the KS equations (i.e., performing a DFT calculation). From now on, we will refer to the DFT k -point grid as k_{DFT} and the Σ k -point grid as k_{Σ} . Since it is empirically known that Σ is more local and therefore tends to converge much faster with respect to the k -point grid than a DFT calculation^{17,42,54}, it is highly efficient to calculate Σ on a relatively coarse k -point grid, and then solve the effective QP equations, in which V_{xc}^{KS} is replaced by Σ_0 interpolated on a fine k -point grid. First, this allows us to decouple the convergence of k_{DFT} and k_{Σ} . Second, GW band structures and other post-processing steps, such as a DOS calculation, can be performed essentially at the cost of a DFT calculation, after Σ is obtained.

We begin by describing the general calculation setup: Using symmetrized structures as described above, we employed the automatic input generator provided with Questaal, `b1m`, to generate input files for the crystal structures in the same manner as Cunningham et al.¹⁷. Unless a parameter is explicitly mentioned below, all parameters were set to their default values generated by `b1m`, which are Questaal defaults in most cases. We generally enable `b1m`'s `gw` flag, generating a larger basis set, better suited for accurate GW calculations. In line with the DeltaCodes project¹⁰⁷, we adjusted the criteria to include more semicore states than the default settings within the basis set to increase the accuracy of the results. Specifically, we included an atomic state as an extended local orbital (ELOs)—local orbitals with a smooth Hankel function tail that extends continuously and differentially into the interstitial region⁴²—in the basis if either the atomic state lies higher than `e1oc` = -2.5 Ry or the share of its corresponding charge beyond the LMTO augmentation sphere is greater than `q1oc` = 0.002. In addition, we included “floating orbitals”⁴² to improve the completeness of the basis set in interstitial regions. These were generated and placed automatically by `b1m`. Questaal employs a threefold representation of wave functions, charge densities, and potentials: a smooth part on a regular mesh, a spherical harmonic expansion within the augmentation spheres, and localized radial functions of the all-electron potential inside the muffin-tin spheres⁴². The smooth part is Fourier transformed to solve the Poisson equation, and the energy cutoff parameter `gmax` indirectly sets the real-space mesh by fixing the maximum Fourier component. The energy cutoff parameter `gmax` used for the DFT calculations is provided by Questaal's `lmfa` code, which is also responsible for building the basis set and computing the free-atom charge densities. It provides two cutoff values by default: one sufficient for accurately describing the valence states and another, larger one that accurately describes the deep core-like states included in the basis as well. This cutoff choice is the all-electron equivalent of the so-called “hardness” used to characterize pseudopotentials in plane-wave pseudopotential codes. We always used the higher value for the benchmark workflow. It is important to note at this point that DFT and GW calculations in Questaal use separate codes, as well as different basis sets and basis set parameters. The GW code of Questaal distinguishes between one- and two-particle plane-wave basis sets. The one-particle basis set describes the interstitial region and is controlled by the energy cutoff parameter `gcutb`, while the two-particle basis set corresponds to the mixed-product basis and is controlled by the cutoff parameter `gcutx`. The cutoff energies for both basis sets were generated using

an internal algorithm included in `b1m`, which sets these cutoffs very conservatively. In other words, it sets them much larger than necessary. To verify the cutoffs used in the GW code of Questaal for our workflow, we compared them to those used by Cunningham et al.¹⁷ for some systems. We found that our cutoff values ($G_{\text{cut}}(Y, M)$ in ref. 17) are consistently higher than the values used there. At last, we adjusted the parameter that controls the previously mentioned interpolation of Σ_0 . Since the high-energy parts of Σ_0 do not interpolate well⁴², Questaal treats them above a cutoff energy Σ_{cut} by removing the off-diagonal elements and replacing the diagonal by an averaged value above Σ_{cut} . As Cunningham et al.¹⁷ have already noted, the results weakly depend on Σ_{cut} . To be safe, we increased the cutoff energy from the default of 2 Ry to 3 Ry (cf. ref. 17, Table XII).

Although automatically generated basis sets are convenient, they can develop pathologies that cause calculations to become unstable. For this reason, we had to manually intervene in our benchmark for compounds containing Hf, Ta, W, and Ti. For Hf, Ta, and W, we exclude the high-lying d orbitals because they interfered with the low-lying $4f$ orbitals. In turn, we added $5p$ orbitals as ELOs to the basis set. For Ta and W, we also added $4f$ orbitals as ELOs. In the case of Hf, we added $5f$ orbitals as high-lying local orbitals everywhere except for HfSe₂, where they caused instabilities in the calculations. For Ti, all local orbitals except the $5d$ ELOs were removed. We limited the maximum angular momentum of orbitals inside the muffin-tin spheres to `lmx` = 4 for all four elements, and we set the angular-momentum cutoff for projecting wave-function tails centered on neighboring sites to `lmxa` = 5 for Ta, W, and Ti and `lmxa` = 6 for Hf. In addition to these manual adjustments to the basis set for these four elements, we had to set `q1oc` back to its default value of 0.005 for AlCuO₂, GaCuO₂, LaCuO₂, ScCuO₂, and Cu₂O in order to obtain stable calculations.

Next, we describe the computational workflow for performing and converging QPG $_0W_0$, QSGW, and QSG $\hat{W}$ calculations automatically using the input settings and basis set described above:

(i) As a first step, we converged the DFT k -point grid k_{DFT} . Similar to our Quantum ESPRESSO workflow, the initial value for the DFT k -point grid was set to a structure-independent reciprocal density of $n_k = 1000$ per atom as defined in `pymatgen`¹⁰⁶. (For CsAu, BaO, and anatase TiO₂, we set the initial density of the DFT k -point grid to $n_k = 1500$.) Then, we converged the total energy change per atom with respect to the k -point grid to less than 10^{-4} Ry, by iteratively increasing the k -point grid in the same manner as described above for step (i) of the Quantum ESPRESSO and Yambo G_0W_0 -PPA workflow. To reduce the computational effort, we performed one iteration of the KS self-consistency cycle per DFT calculation and used the Harris-Foulkes energy¹⁰⁸ per atom as our “total energy” for the k -point grid convergence. We then performed fully self-consistent DFT using the found converged k -point grid, i.e., k_{DFT} .

(ii) In the next step, we converged the self-energy k -point grid k_{Σ} using QPG $_0W_0$ calculations by increasing k_{Σ} in the same manner as k_{DFT} . However, the initial value of the density of the Σ k -point grid was lowered to $n_k = 300$, as Σ tends to converge much faster with respect to the k -point grid than a DFT calculation^{17,42,54}. (For CsAu, BaO, and anatase TiO₂, we set the initial density of the Σ k -point grid to $n_k = 1000$.) We increased k_{Σ} until the difference in QPG $_0W_0$ band gaps was less than 25 meV or k_{DFT} was reached. In rare cases where convergence to below 100 meV was not achieved when k_{DFT} was reached, we omitted the QPG $_0W_0$ and the results of the subsequent calculations for the materials in question from the benchmark dataset. We found that half of the DFT k -point grid density or less was, in most cases, sufficient for convergence, again highlighting the more local character of the self-energy. To obtain the QPG $_0W_0$ +SOC band gap, we performed a post-processing step in which we solved the effective QP equations, including the interpolated Σ_0 and the diagonal parts of $\mathbf{L} \cdot \mathbf{S}$, as described in detail by Pashov et al.⁴² (cf. Section 2.8.2 and Section 3.9).

(iii) Because the subsequent QSGW and QSG $\hat{W}$ steps are much more demanding, we use the penultimate k_{Σ} from step (ii) for evaluating Σ from this step onward. This reduces wall time by nearly an order of magnitude while maintaining the band gap difference within our convergence criterion of 25 meV (or 100 meV in rare cases, see above). Starting from the QPG $_0W_0$

self-energy, we iterated the QSGW cycle until the root mean square (RMS) change in Σ_0 reaches 10^{-5} Ry. We stopped the self-consistency cycle if a calculation did not converge by the 25th iteration. This was done because we noticed that when convergence was difficult to achieve, the RMS tended to oscillate around and below 10^{-4} . However, the significant digits of the band gap did not change during these self-energy oscillations. Fortunately, this has only rarely happened. Also, note that stopping after iteration 25 if convergence is not achieved by then also saves computing resources.

(iv) Starting from the QSGW self-energy, we iterated the QSGW in the same manner as in step (iii), again using the penultimate k_{Σ} from step (ii). The transition space, i.e., the number of valence and conduction bands included in the Bethe–Salpeter Hamiltonian, was set using a heuristic, as converging QSGW calculations for all materials with respect to the BSE transition space is computationally infeasible: We simply included in the Bethe–Salpeter Hamiltonian all bands 10 eV below (above) the valence band maximum (conduction band minimum). Additionally, we include bands more than 10 eV below the valence band maximum as long as each subsequent band is separated from the previous one by less than 0.5 eV. This is done to include all bands from a specific valence band manifold, e.g., some semicore states, in the BSE transition space. For the conduction bands, we used a minimum of eight bands, which was helpful for materials such as Si, C, Ne, and LiF, where the conduction bands are spread over a large energy range (i.e., they have a large dispersion). For materials that were also calculated by Cunningham et al.¹⁷, we compared the obtained BSE transition spaces and found that our heuristic produced transition spaces that were equal to or larger than those used there. Similar to step (iii), the self-consistency cycle did not converge after 25 iterations for a few materials. We observed again that the significant digits of the band gap did not change anymore, and the RMS fluctuated around and below 10^{-4} . Analogous to step (ii), we obtained the QSGW+SOC band gap as a post-processing step.

As a post-processing step after each workflow step, we calculated band structures (including element projections) and DOS for each calculated theory level, except for calculations that include SOC and the α QSGW calculations. All band structures were calculated based on k-point paths taken from ref. 109 as implemented in Questaal. These can be found in the provided database entries, see Code and data availability statement.

To extract band gaps from our all-electron Questaal calculations, we combined two Brillouin zone sampling strategies—(i) a regular three-dimensional k-point grid ($E_{\text{gap}}^{\text{kDFT}}$); (ii) k-point paths along symmetry lines¹⁰⁹ ($E_{\text{gap}}^{\text{ksym}}$), i.e., a band structure calculation—and, for each system, reported:

$$E_{\text{gap}} = \min(E_{\text{gap}}^{\text{kDFT}}, E_{\text{gap}}^{\text{ksym}})$$

For materials whose extrema lie along high-symmetry lines—silicon being the canonical example—a conventional band structure path correctly determines the band gap. In contrast, for materials such as LiCoO₂⁹⁶, the band extrema can be located at general k-points anywhere in the Brillouin zone. Thus, a band structure calculation cannot capture them, and a regular three-dimensional k-point grid is required. However, the average (signed) difference between the two approaches is, in our case, only 4 meV, with a mean absolute difference of 7 meV. Outliers exist, though, such as the aforementioned LiCoO₂⁹⁶, see Supplementary Note 8. Since we did not compute band structures for the QPG₀W₀+SOC, α QSGW, and QPGW+SOC calculations, we omit band gaps for these methods from the benchmark whenever the LDA $E_{\text{gap}}^{\text{kDFT}}$ was more than 25 meV smaller than $E_{\text{gap}}^{\text{kDFT}}$, which occurred only three times.

In conclusion, the automated execution of the complex workflow outlined above—which involves performing and converging all-electron QPG₀W₀ followed by self-consistent QSGW and its vertex-corrected variant, QSGW—for a large set of materials, is a formidable computational challenge. As described above, we acknowledge that a few pragmatic shortcuts were unavoidable. Nevertheless, we are confident that the reported band gaps are converged to an accuracy of about 100 meV or better.

Code and data availability

The high-throughput MBPT workflows used in this study, along with the generated data and post-processing code, are available at https://github.com/MaxGrossmann/mbpt_benchmark or <https://doi.org/10.5281/zenodo.17279454>. The Questaal database entries found in the repository provide numerical data on band structures and density of states for all calculated materials (excluding SOC and α QSGW results) in JSON format. In addition, we also provide a Jupyter notebook that illustrates how to load and visualize the data. Furthermore, we included ready-to-use plots of band structures and density of states at the QPG₀W₀, QSGW, and QSGW levels, organized by material composition, ICSD identifier, and theory level as PDF files.

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References

- Shockley, W. *Electrons and Holes in Semiconductors: With Applications to Transistors* (D. Van Nostrand, 1950).
- Perdew, J. P. Density functional theory and the band gap problem. *Int. J. Quantum Chem.* **28**, 497–523 (2009).
- Borlido, P. et al. Large-scale benchmark of exchange-correlation functionals for the determination of electronic band gaps of solids. *J. Chem. Theory Comput.* **15**, 5069–5079 (2019).
- Borlido, P. et al. Exchange-correlation functionals for band gaps of solids: benchmark, reparametrization and machine learning. *npj Comput. Mater.* **6**, 96 (2020).
- Perdew, J. P. & Schmidt, K. Jacob's ladder of density functional approximations for the exchange-correlation energy. *AIP Conf. Proc.* **577**, 1–20 (2001).
- Hedin, L. New method for calculating the one-particle green's function with application to the electron-gas problem. *Phys. Rev.* **139**, A796–A823 (1965).
- Onida, G., Reining, L. & Rubio, A. Electronic excitations: density-functional versus many-body Green's-function approaches. *Rev. Mod. Phys.* **74**, 601–659 (2002).
- Shishkin, M. & Kresse, G. Implementation and performance of the frequency-dependent GW method within the PAW framework. *Phys. Rev. B* **74**, 035101 (2006).
- Shishkin, M. & Kresse, G. Self-consistent GW calculations for semiconductors and insulators. *Phys. Rev. B* **75**, 235102 (2007).
- Kotani, T., van Schilfgaarde, M. & Faleev, S. V. Quasiparticle self-consistent GW method: a basis for the independent-particle approximation. *Phys. Rev. B* **76**, 165106 (2007).
- Shishkin, M., Marsman, M. & Kresse, G. Accurate quasiparticle spectra from self-consistent GW calculations with vertex corrections. *Phys. Rev. Lett.* **99**, 246403 (2007).
- Liu, P., Kaitak, M., Klimeš, J. & Kresse, G. Cubic scaling GW: towards fast quasiparticle calculations. *Phys. Rev. B* **94**, 165109 (2016).
- Kutepov, A. L. Self-consistent GW method: O(N) algorithm for polarizability and self energy. *Comput. Phys. Commun.* **257**, 107502 (2020).
- Leon, D. A. et al. Frequency dependence in GW made simple using a multipole approximation. *Phys. Rev. B* **104**, 115157 (2021).
- Kutepov, A. L. Full versus quasiparticle self-consistency in vertex-corrected GW approaches. *Phys. Rev. B* **105**, 045124 (2022).
- Leon, D. A., Ferretti, A., Varsano, D., Molinari, E. & Cardoso, C. Efficient full frequency GW for metals using a multipole approach for the dielectric screening. *Phys. Rev. B* **107**, 155130 (2023).
- Cunningham, B., Grüning, M., Pashov, D. & van Schilfgaarde, M. QS G $\dot{W}$: quasiparticle self-consistent GW with ladder diagrams in W. *Phys. Rev. B* **108**, 165104 (2023).
- Butler, K. T., Davies, D. W., Cartwright, H., Isayev, O. & Walsh, A. Machine learning for molecular and materials science. *Nature* **559**, 547–555 (2018).

19. Stanev, V. et al. Machine learning modeling of superconducting critical temperature. *npj Comput. Mater.* **4**, 29 (2018).
20. Cerqueira, T. F. T., Sanna, A. & Marques, M. A. L. Sampling the materials space for conventional superconducting compounds. *Adv. Mater.* **36**, 2307085 (2023).
21. Sanna, A. et al. Prediction of ambient pressure conventional superconductivity above 80 K in hydride compounds. *npj Comput. Mater.* **10**, 44 (2024).
22. Grunert, M. et al. Modeling complex proton transport phenomena—exploring the limits of fine-tuning and transferability of foundational machine-learned force fields. *J. Phys. Chem. C* **129**, 9662–9669 (2025).
23. Angeletti, A. et al. Hydrogen diffusion in magnesium using machine learning potentials: a comparative study. *npj Comput. Mater.* **11**, 85 (2025).
24. Ibrahim, A. & Ataca, C. Prediction of frequency-dependent optical spectrum for solid materials: a multioutput and multifidelity machine learning approach. *ACS Appl. Mater. Inter.* **16**, 41145–41156 (2024).
25. Hung, N. T., Okabe, R., Chottratanapituk, A. & Li, M. Universal ensemble-embedding graph neural network for direct prediction of optical spectra from crystal structures. *Adv. Mater.* **36**, 2409175 (2024).
26. Grunert, M., Großmann, M. & Runge, E. Deep learning of spectra: predicting the dielectric function of semiconductors. *Phys. Rev. Mater.* **8**, L122201 (2024).
27. Hoffmann, N., Schmidt, J., Botti, S. & Marques, M. A. L. Transfer learning on large datasets for the accurate prediction of material properties. *Digit. Discov.* **2**, 1368–1379 (2023).
28. Grunert, M., Großmann, M. & Runge, E. Machine learning climbs the Jacob’s Ladder of optoelectronic properties. *Nat. Commun.* **16**, 1–10 (2025).
29. Bergerhoff, G., Hundt, R., Sievers, R. & Brown, I. D. The inorganic crystal structure data base. *J. Chem. Inf. Comput. Sci.* **23**, 66–69 (1983).
30. Belsky, A., Hellenbrandt, M., Karen, V. L. & Luksch, P. New developments in the inorganic crystal structure database (ICSD): accessibility in support of materials research and design. *Acta Crystallogr. B* **58**, 364–369 (2002).
31. Zagorac, D., Müller, H., Ruehl, S., Zagorac, J. & Rehme, S. Recent developments in the inorganic crystal structure database: theoretical crystal structure data and related features. *J. Appl. Crystallogr.* **52**, 918–925 (2019).
32. Tran, F. & Blaha, P. Accurate band gaps of semiconductors and insulators with a semilocal exchange–correlation potential. *Phys. Rev. Lett.* **102**, 226401 (2009).
33. Heyd, J., Scuseria, G. E. & Ernzerhof, M. Hybrid functionals based on a screened Coulomb potential. *J. Chem. Phys.* **118**, 8207–8215 (2003).
34. Heyd, J., Scuseria, G. E. & Ernzerhof, M. Erratum: “Hybrid functionals based on a screened Coulomb potential” [*J. Chem. Phys.* **118**, 8207 (2003)]. *J. Chem. Phys.* **124**, 219906 (2006).
35. Reining, L. The GW approximation: content, successes and limitations. *WIREs Comput. Mol. Sci.* **8**, e1344 (2018).
36. Godby, R. W., Schlüter, M. & Sham, L. J. Self-energy operators and exchange–correlation potentials in semiconductors. *Phys. Rev. B* **37**, 10159–10175 (1988).
37. Marques, M. A., Oliveira, M. J. & Burnus, T. Libxc: a library of exchange and correlation functionals for density functional theory. *Comput. Phys. Commun.* **183**, 2272–2281 (2012).
38. Giannozzi, P. et al. QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials. *J. Phys. Condens. Matter* **21**, 395502 (2009).
39. Giannozzi, P. et al. Advanced capabilities for materials modelling with Quantum ESPRESSO. *J. Phys. Condens. Matter* **29**, 465901 (2017).
40. Marini, A., Hogan, C., Grüning, M. & Varsano, D. yambo: an ab initio tool for excited state calculations. *Comput. Phys. Commun.* **180**, 1392–1403 (2009).
41. Sangalli, D. et al. Many-body perturbation theory calculations using the Yambo code. *J. Phys.: Condens. Matter* **31**, 325902 (2019).
42. Pashov, D. et al. Questaal: a package of electronic structure methods based on the linear muffin-tin orbital technique. *Comput. Phys. Commun.* **249**, 107065 (2020).
43. Hybertsen, M. S. & Louie, S. G. Electron correlation in semiconductors and insulators: Band gaps and quasiparticle energies. *Phys. Rev. B* **34**, 5390–5413 (1986).
44. Deslippe, J. et al. BerkeleyGW: a massively parallel computer package for the calculation of the quasiparticle and optical properties of materials and nanostructures. *Comput. Phys. Commun.* **183**, 1269–1289 (2012).
45. Mortensen, J. J. et al. GPAW: an open Python package for electronic structure calculations. *J. Chem. Phys.* **160**, 092503 (2024).
46. van Schilfgaarde, M., Kotani, T. & Faleev, S. V. Adequacy of approximations in GW theory. *Phys. Rev. B* **74**, 245125 (2006).
47. Bruneval, F., Sottile, F., Olevano, V., Del Sole, R. & Reining, L. Many-body perturbation theory using the density-functional concept: beyond the GW approximation. *Phys. Rev. Lett.* **94**, 186402 (2005).
48. Franchini, C., Sanna, A., Marsman, M. & Kresse, G. Structural, vibrational, and quasiparticle properties of the Peierls semiconductor BaBiO₃: a hybrid functional and self-consistent GW + vertex-corrections study. *Phys. Rev. B* **81**, 085213 (2010).
49. Tal, A., Chen, W. & Pasquarello, A. Vertex function compliant with the Ward identity for quasiparticle self-consistent calculations beyond GW. *Phys. Rev. B* **103**, L161104 (2021).
50. Kutepov, A. L. Electronic structure of Na, K, Si, and LiF from self-consistent solution of Hedin’s equations including vertex corrections. *Phys. Rev. B* **94**, 155101 (2016).
51. Kutepov, A. L. Self-consistent solution of Hedin’s equations: semiconductors and insulators. *Phys. Rev. B* **95**, 195120 (2017).
52. Jain, A. et al. Commentary: the materials project: a materials genome approach to accelerating materials innovation. *APL Mater.* **1**, 011002 (2013).
53. Ong, S. P. et al. The Materials Application Programming Interface (API): a simple, flexible, and efficient API for materials data based on REpresentational State Transfer (REST) principles. *Comput. Mater. Sci.* **97**, 209–215 (2015).
54. Großmann, M., Grunert, M. & Runge, E. A robust, simple, and efficient convergence workflow for GW calculations. *npj Comput. Mater.* **10**, 135 (2024).
55. Tofallis, C. A better measure of relative prediction accuracy for model selection and model estimation. *J. Oper. Res. Soc.* **66**, 1352–1362 (2015).
56. Carriço, P. J. M. A., Ferreira, M., Cerqueira, T. F. T., Nogueira, F. & Borlido, P. High-refractive-index materials screening from machine learning and ab initio methods. *Phys. Rev. Mater.* **8**, 015201 (2024).
57. Bohm, D. & Pines, D. A collective description of electron interactions. I. Magnetic interactions. *Phys. Rev.* **82**, 625–634 (1951).
58. Gant, S. E. et al. Optimally tuned starting point for single-shot GW calculations of solids. *Phys. Rev. Mater.* **6**, 053802 (2022).
59. Perdew, J. P., Ernzerhof, M. & Burke, K. Rationale for mixing exact exchange with density functional approximations. *J. Chem. Phys.* **105**, 9982–9985 (1996).
60. Deguchi, D., Sato, K., Kino, H. & Kotani, T. Accurate energy bands calculated by the hybrid quasiparticle self-consistent GW method implemented in the ecalj package. *Jpn. J. Appl. Phys.* **55**, 051201 (2016).
61. Bhandari, C., van Schilfgaarde, M., Kotani, T. & Lambrecht, W. R. L. All-electron quasiparticle self-consistent GW band structures for SrTiO₃ including lattice polarization corrections in different phases. *Phys. Rev. Mater.* **2**, 013807 (2018).

62. Chantis, A. N., van Schilfgaarde, M. & Kotani, T. Ab initio prediction of conduction band spin splitting in zinc blende semiconductors. *Phys. Rev. Lett.* **96**, 086405 (2006).
63. Madelung, O. *Semiconductors: Data Handbook* (Springer, Heidelberg, 2003).
64. Marchetti, A. P. & Bottger, G. L. Optical absorption spectrum of AgF. *Phys. Rev. B* **3**, 2604–2607 (1971).
65. Koriche, N., Bouguelia, A. & Trari, M. Photocatalytic hydrogen production over new oxide CuLaO_{2.62}. *Int. J. Hydrogen Energy*. **31**, 1196–1203 (2006).
66. Li, S. et al. Preparation and characterization of solution-processed nanocrystalline p-type CuAlO₂ thin-film transistors. *Nanoscale Res. Lett.* **13**, 259 (2018).
67. Lukashev, P., Lambrecht, W. R. L., Kotani, T. & van Schilfgaarde, M. Electronic and crystal structure of Cu_{2–x}S: Full-potential electronic structure calculations. *Phys. Rev. B* **76**, 195202 (2007).
68. Råsaender, M., Bergqvist, L. & Delin, A. Density functional theory study of the electronic structure of fluorite Cu₂Se. *J. Condens. Matter Phys.* **25**, 125503 (2013).
69. Zhang, Y. et al. Electronic structure of antiferromagnetic Cu₂X (X = S, Se, Te) within the modified Becke–Johnson potential plus an on-site Coulomb *U*. *J. Chem. Phys.* **140**, 074702 (2014).
70. Klan, J. M. et al. Theoretical study of the impact of vacancies and disorder on the electronic properties of Cu_{2–x}Se. *J. Phys. Chem. C* **125**, 12324–12332 (2021).
71. Yan, Z., Perdew, J. P. & Kurth, S. Density functional for short-range correlation: accuracy of the random-phase approximation for isoelectronic energy changes. *Phys. Rev. B* **61**, 16430–16439 (2000).
72. Acharya, S. et al. A theory for colors of strongly correlated electronic systems. *Nat. Commun.* **14**, 5565 (2023).
73. Aycibin, M. et al. Physical properties of RbAu compound. *Comput. Condens. Matter* **1**, 32–37 (2014).
74. Nicoloso, N. & Freyland, W. Magnetic susceptibility and optical absorption studies of Rb-Au alloys. *Ber. Bunsenges. Phys. Chem.* **88**, 953–956 (1984).
75. Liu, T. L. Optical properties of RbAu and CsAu. *Phys. Rev. B* **12**, 3008–3012 (1975).
76. Martin, F. E. & Hensley, E. B. Photoelectric emission from Barium telluride. *Phys. Rev.* **163**, 219–223 (1967).
77. Wilmers, K. et al. Ellipsometric studies of Be_xZn_{1–x}Se between 3 eV and 25 eV. *Phys. Rev. B* **59**, 10071–10075 (1999).
78. Eich, D. et al. Resonant inelastic soft X-ray scattering of Be chalcogenides. *Phys. Rev. B* **73**, 115212 (2006).
79. Grunert, M., Großmann, M. & Runge, E. Predicting exciton binding energies from ground-state properties. *Phys. Rev. B* **110**, 075204 (2024).
80. McLeod, J. A. et al. Band gaps and electronic structure of alkaline-earth and post-transition-metal oxides. *Phys. Rev. B* **81**, 245123 (2010).
81. Saum, G. A. & Hensley, E. B. Fundamental optical absorption in the IIa-VIb compounds. *Phys. Rev.* **113**, 1019–1022 (1959).
82. Kaneko, Y. & Koda, T. New developments in IIa-VIb (alkaline-earth chalcogenide) binary semiconductors. *J. Cryst. Growth* **86**, 72–78 (1988).
83. Förster, A. & Bruneval, F. Why does the GW approximation give accurate quasiparticle energies? The cancellation of vertex corrections quantified. *J. Phys. Chem. Lett.* **15**, 12526–12534 (2024).
84. Förster, A. Beyond quasi-particle self-consistent GW for molecules with vertex corrections. *J. Chem. Theory Comput.* **21**, 1709–1721 (2025).
85. Cardona, M. Electron-phonon interaction in tetrahedral semiconductors. *Solid State Commun.* **133**, 3–18 (2005).
86. Giustino, F., Louie, S. G. & Cohen, M. L. Electron-phonon renormalization of the direct band gap of diamond. *Phys. Rev. Lett.* **105**, 265501 (2010).
87. Monserrat, B. & Needs, R. J. Comparing electron-phonon coupling strength in diamond, silicon, and silicon carbide: first-principles study. *Phys. Rev. B* **89**, 214304 (2014).
88. Sugiura, C. *k*-x-ray spectra and electronic band structures of MgCl₂, CaCl₂, SrCl₂, and BaCl₂. *Phys. Rev. B* **9**, 2679–2683 (1974).
89. Taylor, A. J., Gibson, R. B. & Roberts, J. P. Two-photon absorption at 248 nm in ultraviolet window materials. *Opt. Lett.* **13**, 814 (1988).
90. Kinno, S. & Onaka, R. The optical properties and the band structure of magnesium chloride. *J. Phys. Soc. Jpn.* **49**, 1379–1386 (1980).
91. Rubloff, G. W. Far-ultraviolet reflectance spectra and the electronic structure of ionic crystals. *Phys. Rev. B* **5**, 662–684 (1972).
92. Findley, P. R., Wu, Z. & Walker, W. C. Temperature dependence of the electronic band gap and exciton states of superionic PbF₂. *Phys. Rev. B* **28**, 4761–4767 (1983).
93. Al-Brithe, H. A., Smith, A. R. & Gall, D. Surface and bulk electronic structure of ScN(001) investigated by scanning tunneling microscopy/spectroscopy and optical absorption spectroscopy. *Phys. Rev. B* **70**, 045303 (2004).
94. Deng, R., Ozsdolay, B. D., Zheng, P. Y., Khare, S. V. & Gall, D. Optical and transport measurement and first-principles determination of the ScN band gap. *Phys. Rev. B* **91**, 045104 (2015).
95. Grümbel, J. et al. Band gaps and phonons of quasi-bulk rocksalt ScN. *Phys. Rev. Mater.* **8**, L071601 (2024).
96. Radha, S. K. et al. Optical response and band structure of LiCoO₂ including electron-hole interaction effects. *Phys. Rev. B* **104**, 115120 (2021).
97. Großmann, M. et al. On the origin of bulk-related anisotropies in surface optical spectra. *arXiv:2503.11253* (2025).
98. Flötotto, A., Beenken, W. J. D., Lauer, K., Krischok, S. & Runge, E. Electronic structure and energy landscape of B₅₇Si₁ related defects. *Phys. Rev. Mater.* **9**, 024601 (2025).
99. Aouina, A., Borlido, P., Marques, M. A. L. & Botti, S. Assessing exchange-correlation functionals for accurate densities of solids. *J. Chem. Theory Comput.* **20**, 10852–10860 (2024).
100. Lee, H. et al. Electron-phonon physics from first principles using the EPW code. *npj Comput. Mater.* **9**, 156 (2023).
101. Li, Z., Antonius, G., Chan, Y.-H. & Louie, S. G. Electron-phonon coupling from GW perturbation theory: Practical workflow combining BerkeleyGW, ABINIT, and EPW. *Comput. Phys. Commun.* **295**, 109003 (2024).
102. Togo, A. & Tanaka, I. Spglib: a software library for crystal symmetry search. *arXiv:1808.01590* (2018).
103. Hamann, D. R. Optimized norm-conserving Vanderbilt pseudopotentials. *Phys. Rev. B* **88**, 085117 (2013).
104. van Setten, M. et al. The PseudoDojo: training and grading a 85 element optimized norm-conserving pseudopotential table. *Comput. Phys. Commun.* **226**, 39–54 (2018).
105. Bruneval, F. & Gonze, X. Accurate GW self-energies in a plane-wave basis using only a few empty states: towards large systems. *Phys. Rev. B* **78**, 085125 (2008).
106. Ong, S. P. et al. Python Materials Genomics (pymatgen): a robust, open-source Python library for materials analysis. *Comput. Mater. Sci.* **68**, 314–319 (2013).
107. Lejaeghere, K. et al. Reproducibility in density functional theory calculations of solids. *Science* **351**, aad3000 (2016).
108. Foulkes, W. M. C. & Haydock, R. Tight-binding models and density-functional theory. *Phys. Rev. B* **39**, 12520–12536 (1989).
109. Hinuma, Y., Pizzi, G., Kumagai, Y., Oba, F. & Tanaka, I. Band structure diagram paths based on crystallography. *Comput. Mater. Sci.* **128**, 140–184 (2017).
110. Bernstorff, S. & Saile, V. Experimental determination of band gaps in rare gas solids. *Opt. Commun.* **58**, 181–186 (1986).
111. Roessler, D. & Walker, W. Electronic spectrum of crystalline lithium fluoride. *J. Phys. Chem. Solids* **28**, 1507–1515 (1967).

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Author contributions

M.G. and M.G. conceived the idea; M.T. wrote the workflow and performed all the computations using the Quantum ESPRESSO and Yambo codes; M. Großmann wrote the workflow and carried out all calculations using the Questaal code; M. Grunert optimized the LMTO basis sets in Questaal for materials containing Hf, Ta, W, and Ti; M.T. and M. Großmann analyzed the data and visualized the results; M. Großmann wrote the first draft of the manuscript; E.R. supervised the work; all authors revised and approved the manuscript.

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Correspondence and requests for materials should be addressed to Max Großmann.

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